Geophysics IIIA: Week 2

Scenarios: Decision-Making for
Physical Property Mixing

Instructions: Work in mixed geology/physics groups. For each scenario: (1) state the *goal* and *key properties*; (2) choose how you will *obtain* those properties (measure, estimate from composition, infer from proxies); (3) select an *appropriate mixing law* (justify); (4) identify *uncertainties/non-uniqueness*; (5) draft **2–3 questions a geophysicist should ask a geologist** before committing to a survey/model. Use the note space provided.

Scenario 1 — Gravity over Mixed Lithologies

You are planning a gravity survey over shallow basement with weathered granite, mafic dyke swarms, and variably porous sediments.
Tasks:

1. Which density components dominate the expected anomaly? When can density be predicted compositionally? When does porosity matter most?
2. Choose a mixing law for sediments vs. crystalline units.
3. Note assumptions and unit conventions.
4. List major uncertainty sources and how you would reduce them.
5. Write 2–3 questions for the geologist.

Scenario 2 — Magnetic Lineament Interpretation

You have regional aeromagnetic data (20 km line spacing) over volcaniclastics and granitoids.
Tasks:

1. Identify which property matters (susceptibility vs. remanence) and what fabric/mineralogy might control it.
2. Would you estimate χ from lithology or require measurements? What are the risks of assuming induced-only magnetization?
3. What geological information (e.g., thermal/alteration history) would change your interpretation most?
4. Write 2–3 questions for the geologist.

Scenario 3 — Conductive Heat-Flow Study

You need estimates of thermal conductivity $k(T)$ and heat production A for three lithologies to model a geotherm.

Tasks:

1. Select mixing laws for k in random vs. layered media; justify.
2. Decide how to estimate A (e.g., from K, U, Th or lithology) and how uncertainty propagates into geotherms.
3. Identify which additional measurements would reduce uncertainty most.
4. Write 2–3 questions for the geologist.

Scenario 4 — Seismic Property Estimation

You must propose a 1-D velocity model.
Tasks:

1. Explain why Voigt–Reuss–Hill bounds are appropriate for moduli/velocities.
2. Distinguish density effects from moduli effects; identify when cracks/porosity dominate.
3. What rock fabric information would you request from geology to constrain anisotropy?
4. Write 2–3 questions for the geologist.

Score Sheet (Scenario Responses)

Student/Group: ________________________ **Scenario(s):** ____________________

Criterion	Description	Points
Property strategy	Chooses appropriate properties and acquisition/estimation strategy with sound assumptions	6
Mixing-law choice	Selects and justifies a law suitable for geometry and property	6
Uncertainty	Identifies and prioritizes key uncertainties; proposes mitigation	5
Geo questions	High-value, concrete questions for the geologist	5
Clarity	Organized, legible, concise notes	3
Total		**25**